

structure, or style) while preserving its factual consistency with the source document. In principle, a robust factuality metric should be invariant to such benign edits, assigning similar scores to the original and perturbed versions.

Following Ramprasad and Wallace (2024), we use seven perturbation types, each targeting a different linguistic dimension. These include *Paraphrased*, where the summary is rewritten with alternate phrasings and syntactic structures; *Simplified*, where complex or compound constructions are rewritten into shorter, more readable sentences; *Synonym Replaced*, where content words are substituted with close synonyms to test for lexical invariance. We also generate *Less Diverse* summaries that reduce vocabulary variation, exploring whether metrics implicitly reward stylistic richness. Additional perturbations include *Negated*, which introduces logically equivalent negations to probe sensitivity to syntactic polarity, *Summarized*, which further compresses the summary to test how conciseness is handled, and *Added Source Text*, which inserts a factual sentence directly from the source that is unrelated to the main summary content. All seven perturbed summaries are generated using the GPT-4o (Hurst et al., 2024) model via the OpenAI API. The detailed prompts used to generate each perturbation are provided in App. A. To ensure that these perturbations preserve factual consistency, we additionally perform an NLI-based faithfulness check comparing each perturbed summary against its original counterpart; detailed results are reported in App. C.

While these transformations are meaning-preserving, they pose particular challenges in the long-document setting: summaries must capture information scattered across thousands of tokens, so perturbations that change sentence structure, reduce vocabulary, or alter flow can disrupt long-range dependencies and retrieval alignment. This makes them a rigorous test of whether factuality metrics remain robust. Any significant fluctuation in scores, despite no factual errors being introduced, indicates that a metric is reacting to surface-level edits rather than faithfully assessing factual consistency.

3.2 Retrieval-Based Scoring for Long Documents

Most factuality metrics in existing literature are designed for short inputs and cannot directly process the full content of long documents due to token

length limitations. This is particularly problematic in long-form summarization, where summaries may draw on information scattered across multiple sections or even multiple documents. To address this, we follow the retrieval-augmented strategy proposed by Bishop et al. (2024), which enables factuality evaluation at the sentence level without requiring the metric to ingest the entire source document at once.

Let $S = \{s_1, s_2, \dots, s_m\}$ denote the summary, where each s_j is a sentence, and let $D = \{d_1, d_2, \dots, d_n\}$ be the set of sentences in the source document. For each summary sentence s_j , we compute a sentence embedding e_j , and similarly obtain embeddings $\{e_1^D, \dots, e_n^D\}$ for the source document using a pre-trained sentence encoder. We compute cosine similarity² between s_j and each $d_i \in D$ and retrieve the top- K most similar source sentences. Each retrieved sentence $d_{j,k}$ is then expanded to include the surrounding context within a symmetric window size w , forming a snippet:

$$d_{j,k}^{(w)} = \{d_{j,k-w}, \dots, d_{j,k}, \dots, d_{j,k+w}\}. \quad (1)$$

We evaluate the factual consistency of s_j against each of the K context snippets, using any automated metric $\mathcal{M}$, and take the maximum score:

$$\text{score}(s_j) = \max_{k \in \{1, \dots, K\}} \mathcal{M}(s_j, d_{j,k}^{(w)}). \quad (2)$$

Finally, the summary-level factuality score is computed by averaging these sentence-level scores across all sentences in the summary. In our experiments, we explore how varying w affects metric behavior, shedding light on the sensitivity of different metrics to retrieval context size.

4 Experimental Setup

Metrics We evaluate six reference-free factuality metrics that represent diverse architectures and scoring paradigms. BARTScore (Yuan et al., 2021) estimates the log-likelihood of the summary given the source using a pretrained BART model³, treating factuality as a conditional generation problem. For consistency with other metrics, we exponentiate the log-likelihood scores to obtain normalized values that are directly comparable across metrics.

²We use SBERT (bert-base-nli-mean-tokens) (Reimers and Gurevych, 2019) for computing all sentence similarities.

³<https://huggingface.co/facebook/bart-large-cnn>

SummaC-Conv and SummaC-ZS (Laban et al., 2022) represent entailment-based approaches that apply pretrained NLI models to compute consistency between summary and source sentence pairs; the former uses a learned aggregation layer, while the latter relies on zero-shot averaging. AlignScore (Zha et al., 2023) leverages contrastive alignment learning across NLI, QA, and summarization tasks and demonstrates strong cross-domain generalization. UniEval (Zhong et al., 2022) formulates the evaluation as a multi-dimensional question answering task within a unified T5-based framework, jointly considering factuality, coherence, relevance, and fluency. Finally, MiniCheck (Tang et al., 2024) is a lightweight, sentence-level factuality classifier that achieves near GPT-4 performance at a fraction of the cost. We use the Bespoke-MiniCheck-7B variant, which ranks highest on the LLM-AggreFact benchmark (Tang et al., 2024), and take advantage of its 32k-token context window to provide full-document context during evaluation. We evaluate these metrics in their publicly released form, reflecting common practice in prior studies that apply summarization metrics without task-specific adaptation (Huang et al., 2021; Yang and Wan, 2022; Sotudeh et al., 2021; Guo et al., 2022).

Datasets We conduct our analysis on three long-document summarization datasets spanning diverse domains: *SQuALITY* (Wang et al., 2022), *LexAbSumm* (T.y.s.s. et al., 2024), and *ScholarQABench* (Asai et al., 2024). *SQuALITY* consists of public domain science fiction stories paired with expert-written summaries that balance narrative abstraction and fine-grained detail. *LexAbSumm* contains legal judgments from the European Court of Human Rights, where summaries are aspect-specific and demand precise distillation of dense legal arguments. *ScholarQABench* is a multi-document benchmark based on open-access computer science papers, where the task is to generate detailed, factual answers to expert-written queries using evidence from multiple documents. We selected these three distinct datasets to ensure broad cross-domain coverage, given their substantial differences in structure, style, and language. Detailed dataset statistics are provided in App. B.

Experimental Design To evaluate metric robustness, we construct perturbed versions of the summaries from each dataset using the seven meaning-preserving transformations described in § 3.1. These edits allow us to test whether metrics ex-

hibit sensitivity to benign changes. For each summary, original and perturbed, we use the framework (§ 3.2) to retrieve source evidence and compute factuality scores using all six metrics discussed above. Beyond this robustness analysis, we also investigate how retrieval granularity and claim information density influence metric behavior. We vary the evidence window size w from Eq. 1 to test how broader or narrower retrieval contexts affect metric performance. Additionally, we measure the information density of each summary sentence using the mean pairwise cosine similarity between its embedding and all sentences in the source document. High information density indicates claims that semantically overlap with many parts of the source and are therefore harder to verify, while low-density claims correspond to specific statements with localized evidence. This analysis reveals how metrics respond to different levels of semantic compression, providing insight into their sensitivity to claim complexity in long-form summarization.

5 Results & Analysis

We evaluate the robustness and behavior of six reference-free factuality metrics across a range of semantic-preserving perturbations, varying retrieval context windows, and differences in claim information density. Our results are presented in three parts: (1) robustness under perturbation (§ 5.1), (2) sensitivity to retrieval granularity (§ 5.2), and (3) metric sensitivity to claim information density (§ 5.3).

5.1 Robustness Against Perturbations

Fig. 2 presents the change in factuality scores when different perturbations are applied to the summaries across three datasets. Each plot shows the difference in score (perturbed minus original) for a given metric, broken down by perturbation and dataset. A robust metric should remain invariant to these semantic-preserving changes. However, we find that all metrics show varying levels of sensitivity.

On *LexAbSumm*, BARTScore shows clear negative shifts across nearly all perturbations, while remaining relatively consistent on all other datasets. These consistent declines on *LexAbSumm* indicate that BARTScore is highly sensitive to even mild surface-level edits in the legal domain, where long and complex sentence structures and domain-specific jargon likely amplify generation-based instability. MiniCheck shows very small changes

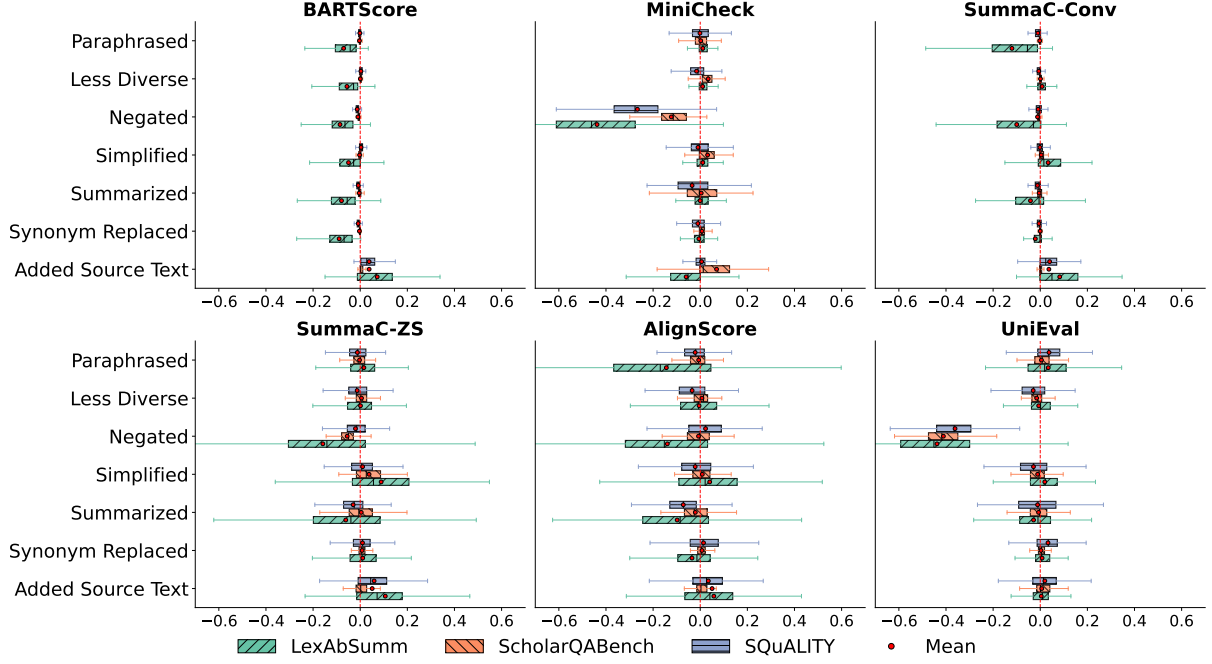


Figure 2: Score change under factuality-preserving perturbations. Boxplots show the difference in factuality score between the perturbed and original summaries, for each metric and perturbation type, across three datasets. The central dot indicates the mean score difference, and the whiskers represent the minimum and maximum values.

across all perturbations and datasets. However, it struggles with logically equivalent negations, especially in *LexAbSumm*. This may reflect a domain mismatch, as the metric appears less effective in capturing factual consistency in legal texts. SummaC-Conv and SummaC-ZS, both based on NLI, show moderate and more balanced behavior. They are somewhat affected by *Summarized* and *Negated* summaries, especially on *SQuALITY* and *LexAbSumm*, showing they are not fully invariant to meaning-preserving rewrites. UniEval is sensitive to most of the perturbations. However, it consistently fails to handle logically equivalent *Negated* summaries across all datasets, suggesting a lack of sensitivity to logical form. AlignScore mostly struggles with the legal domain, showing large score drops in response to *Paraphrased*, *Negated* and *Summarized* summaries. This suggests difficulty in tracking sentence order and logical consistency in structured, formal texts. While it performs more reliably on *SQuALITY* and *ScholarQABench*, it remains less robust than other metrics overall.

Detailed per-dataset results are provided in App. D. These results list the mean factuality scores for each metric and perturbation type across all datasets. To quantify domain-specific instability that is masked by signed averages, we also analyze mean absolute score changes under perturbations;

detailed results are provided in App. E.

5.2 Effect of Retrieval Context Window Size

Table 1 reports average factuality scores on the original summaries for each metric and dataset, using window sizes $w = 0, 1, 2$ in Eq. 1. We find that most metrics show consistent improvements as the window size increases, suggesting that they can effectively leverage broader local context when making sentence-level factuality judgments. This pattern is particularly pronounced on *LexAbSumm*, where understanding legal arguments often requires attending to multi-sentence spans. SummaC-ZS and SummaC-Conv display little sensitivity to larger context windows, implying that their underlying models base judgments on more localized comparisons and are less responsive to extended evidence.

Taken together, these results indicate that retrieval-based scoring can improve factuality assessment in long-document summarization, especially when a broader context is provided. However, NLI-based metrics remain insensitive to increasing context windows.

5.3 Metric Sensitivity to Claim Similarity

To understand what makes factual consistency evaluation difficult in long documents, we analyze how the semantic density of a claim relates to met-